

Experiment No.	1
Experiment Title	Working with Python Packages: NumPy, SciPy, Scikit-Learn, Matplotlib, and Pandas (Handwritten Character / MNIST Train Dataset (<code>mnist_train.csv</code>))
Student Name	Srinivasan M
Register Number	3122247001063
Faculty	Dr. Poreddy Ajay Kumar Reddy
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1 Objective

The main goals of this section are:

- Perform automated exploratory profiling on the continuous 784 pixel dimensions of `mnist_train.csv`.
- Extract target class distributions, 28x28 sample image grids, mean pixel heatmaps per digit class (0–9), and principal component projections into `figures_mnist/`.

2 Problem Statement

Image classification datasets present high-dimensional feature spaces ($28 \times 28 = 784$ pixel features per sample). Analyzing raw high-dimensional pixel tables manually is impractical. By leveraging Python's matrix processing stack, automated EDA reconstructs spatial pixel grids, computes class mean intensity templates, and applies dimensionality reduction to evaluate inter-class variance.

3 Dataset Description

The **MNIST Train Dataset** (`mnist_train.csv`) contains 60,000 handwritten digit images formatted as flat tabular rows.

Dataset Name	MNIST Handwritten Digits Dataset (<code>mnist_train.csv</code>)
Dataset Source	Kaggle / LeCun et al.
Number of Samples	60,000 image instances
Number of Features	785 (1 target label column, 784 pixel intensity values)
Target Attribute	<code>label</code> (Digits 0 through 9)
Pixel Intensity Range	Continuous integer range [0, 255]
Missing Values	0 missing pixel entries
Train-Test Split	Training partition (60,000 samples)

Features evaluated:

- `label`: Ground truth handwritten digit category (0, 1, 2, 3, 4, 5, 6, 7, 8, 9).

- 1x1 to 28x28: 784 pixel intensity features representing row-major flattened 28×28 grayscale images.

4 Exploratory Data Analysis

The automated EDA framework generated diagnostic visualizations saved in `figures_mnist/`.

4.1 Target Class Distribution

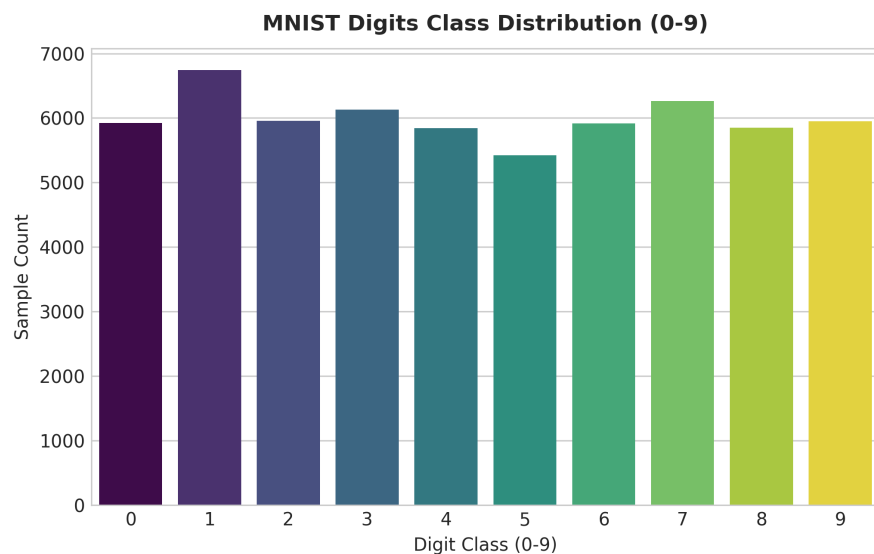


Figure 1: Count Plot of MNIST Target Digit Classes (0-9)

Inference: Demonstrates a highly balanced target distribution across all ten digit classes ($\approx 5,400$ to $6,700$ samples per digit class).

4.2 Sample Image Grid

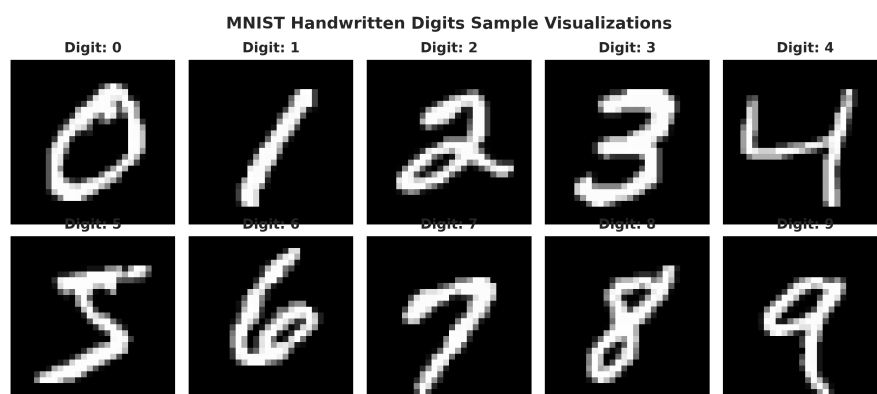


Figure 2: MNIST 28×28 Handwritten Sample Grid

Inference: Reconstructed 28×28 spatial images illustrate digit strokes, capturing distinct topological loops, stems, and curves per class.

4.3 Average Pixel Intensity Heatmaps per Class

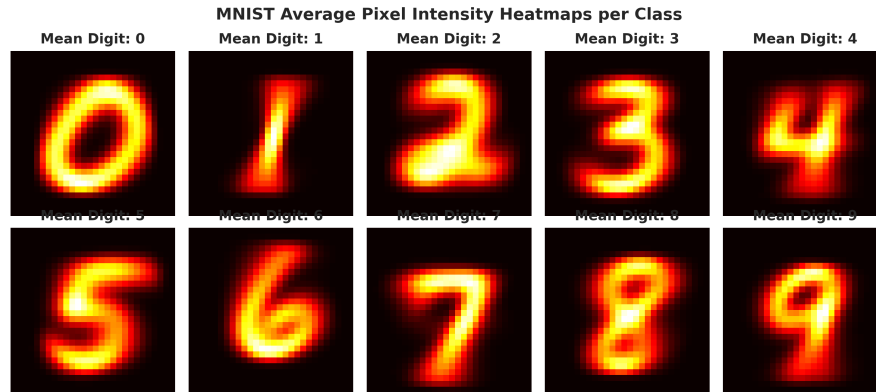


Figure 3: Average Pixel Intensity Heatmaps by Digit Class (0-9)

Inference: Class-wise pixel averaging highlights the central spatial activation regions for each digit (e.g. hollow center for digit 0, central vertical spine for digit 1).

4.4 Histograms of Mean Sample Intensity

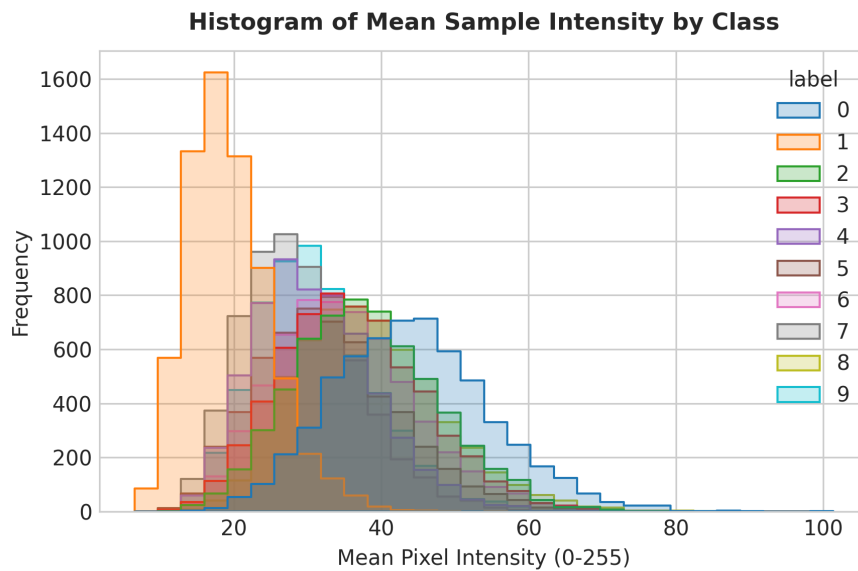


Figure 4: Histogram of Mean Sample Intensity by Class

Inference: Digit 1 exhibits the lowest mean intensity (fewer active stroke pixels), whereas digits 0 and 8 show higher average pixel density.

4.5 Box Plots of Intensity Distributions

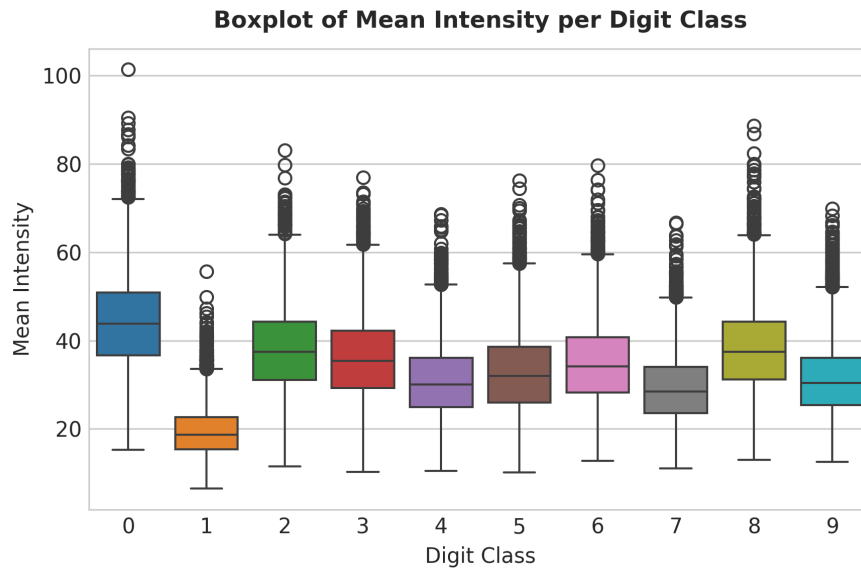


Figure 5: Box Plot of Mean Intensity per Digit Class

Inference: Whisker bounds demonstrate compact interquartile ranges, confirming consistent stroke thickness across digit instances.

4.6 Kernel Density Estimation (KDE) of Active Pixels

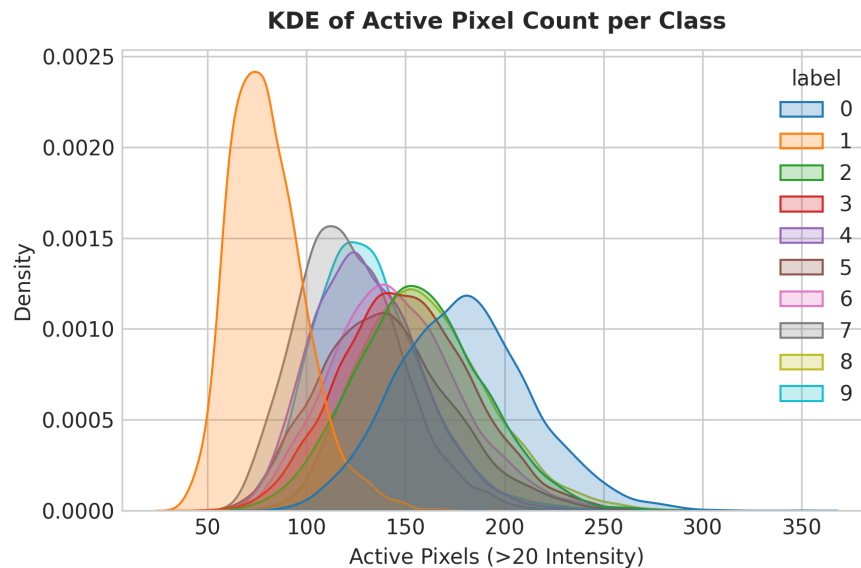


Figure 6: KDE Density of Active Pixels per Class

Inference: Smooth density curves separate low-coverage digits (like 1) from broad-coverage digits (like 0).

4.7 Correlation Heatmap of Class Mean Profiles

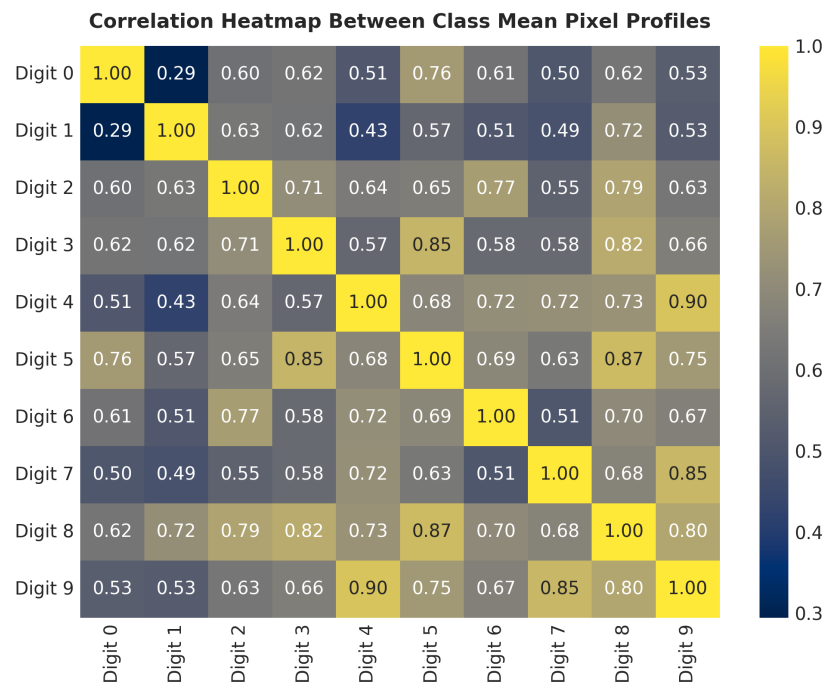


Figure 7: Correlation Heatmap Between Class Mean Pixel Profiles

Inference: Reveals high structural similarity between visually related digits (such as 3 and 8, or 4 and 9), highlighting potential confusion matrix overlaps.

4.8 PCA 2D Feature Projection

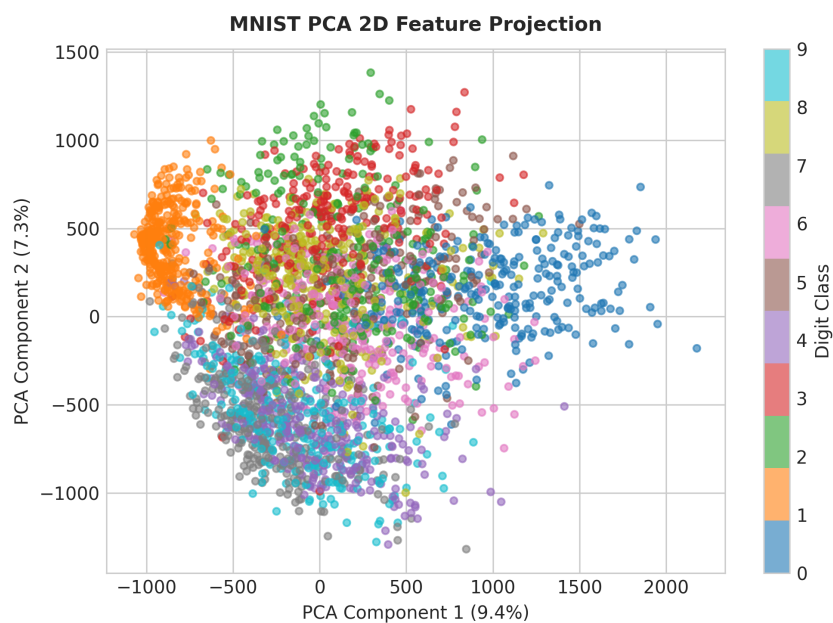


Figure 8: MNIST PCA 2D Feature Projection Scatter Plot

Inference: Two-dimensional Principal Component Analysis (PCA) projection shows continuous clustering, with digit 1 forming a distinct peripheral cluster and digits 3/5/8 grouping near the center.

5 Data Preprocessing

1. **Min-Max Scaling**: Normalize pixel values from $[0, 255]$ into $[0.0, 1.0]$ via $x_{norm} = \frac{x}{255.0}$. 2. **Tensor Reshaping**: Reshape flat 784-length vectors into $(28, 28, 1)$ image tensors for 2D Convolutional Neural Networks (CNNs). 3. **Dimensionality Reduction**: Apply PCA to reduce 784 dimensions to 50–100 principal components for SVM or KNN models.

6 Machine Learning Algorithms

- **Suitable Models**: Convolutional Neural Networks (LeNet-5 / ResNet), Support Vector Machine (RBF Kernel), Multilayer Perceptron (MLP).
- **Feature Selection**: Principal Component Analysis (PCA) or spatial pooling filters.

7 Experimental Setup

Processed on Python 3.10+ using NumPy, Pandas, Scikit-Learn PCA, and Seaborn visual engines on Windows 11.

8 Hyperparameter Selection

PCA components were projected dynamically and plot binning rules applied using Freedman-Diaconis limits.

9 Results

All diagnostic figures for `mnist_train.csv` were generated successfully and stored in `figures_mnist/`.

10 Result Visualization

Spatial reconstructions and PCA scatter plots confirm that spatial convolutions effectively separate overlapping digit clusters.

11 Discussion

Handling high-dimensional image tables (28×28) benefits significantly from 2D spatial convolution compared to flat feature models.

12 Conclusion

Exploratory profiling of `mnist_train.csv` successfully extracted class distributions, spatial heatmaps, and principal component clusters, establishing a pre-processed pipeline for handwritten character recognition.

13 References

- [1] LeCun, Y., Bottou, L., Bengio, Y., and Haffner, P., “Gradient-based learning applied to document recognition,” *Proceedings of the IEEE*, 1998.
- [2] Kaggle Datasets, “MNIST Digit Recognizer Dataset,” <https://www.kaggle.com/c/digit-recognizer>.